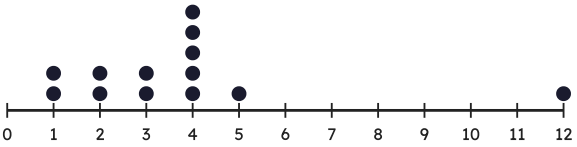
Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____ Calculator: yes

Learning goal: I can choose and justify measures of center and spread. (circle your level after scoring)

4	I can explain how an outlier changes the mean, median, and range.
3	I can choose the best measure of center or spread for a data set and explain why.
2	I can find the mean, median, and range of a data set.
1	I need help to start this benchmark.

1. The line plot shows the number of pets in each household surveyed.

Which measure of center best describes a typical value in this data set, and why?



A. the median, because one value is far from the rest and pulls the mean away

B. the mean, because every value is close to the rest, with no outlier

C. the range, because it shows how spread out the data is

D. the mean, because it is easier to find on a line plot than the median

2. The numbers of minutes a student practiced piano on six days are 32, 41, 50, 57, 64, 186.

Which measure of variation better describes the spread of the data, and why?

A. the range, because there is no outlier to stretch it

B. the mean, because it uses every value in the data set

C. the interquartile range, because an outlier stretches the range

D. the range, because an outlier makes the spread easier to see

3. The prices of the cars in a lot, in thousands of dollars, are 25, 29, 39, 42, 53, 243.

Which measure of center best describes a typical value in this data set?

A. the mean, because no value is far enough away to pull it off center

B. the range, because it shows how spread out the data is

C. the mean, because only the middle value affects it

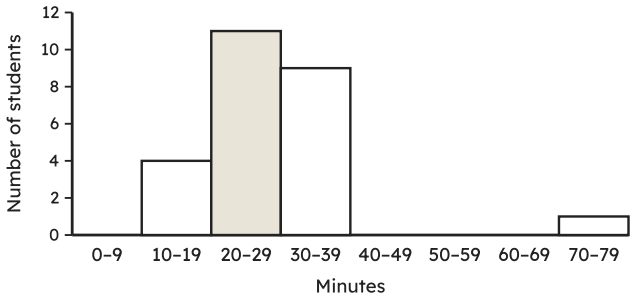
D. the median, because an outlier pulls the mean away from the rest

Turn over →

Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____

4. The histogram shows the numbers of minutes 25 students spent on homework last night.

Which measure of center best describes a typical value in this data set, and why?



A. the range, because one value is far from the rest of the data

B. the mean, because no value is far from the rest of the data

C. the median, because one value is far from the rest of the data

D. the mean, because one value is far from the rest of the data

5. The prices, in dollars, of eight used bikes at a yard sale are 90, 30, 34, 26, 28, 22, 32, 30.

One value in this data set is an outlier. Select **all** the statements that are true.

Select **all** the correct answers.

A. ☐ Removing 90 from the data set would change the mean more than it would change the median.

B. ☐ Removing 90 from the data set would not change the range.

C. ☐ The range is more than twice the interquartile range.

D. ☐ The mean is greater than the median.

E. ☐ The interquartile range is more than half of the range.

F. ☐ Removing 90 from the data set would change the median more than it would change the mean.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.1 · MA.7.DP.1.1 · Choose a measure of center or variation

Unit 13: Numerical and Categorical Data. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A. the median, because one value is far from the rest and pulls the mean away**

One dot sits at 12, far to the right of the cluster between 1 and 5.
That one value pulls the mean toward it, but it does not move the middle value, so the median describes a typical value better.

2. Regular **C. the interquartile range, because an outlier stretches the range**

186 is far from every other value, so it is an outlier.
The range, $186 - 32 = 154$ minutes, is stretched by the outlier. The interquartile range uses only the middle half of the ordered data, $Q_3 - Q_1 = 64 - 41 = 23$ minutes, so the outlier does not stretch it.
The interquartile range describes the spread better; it goes with the median as the measure of center.

3. Regular **D. the median, because an outlier pulls the mean away from the rest**

243 is far greater than every other value, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

4. Regular **C. the median, because one value is far from the rest of the data**

One value, in the 70–79 interval, is separated from all the others by three empty intervals, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

5. Rigor **A. Removing 90 from the data set would change the mean more than it would change the median; C. The range is more than twice the interquartile range; D. The mean is greater than the median.**

In order: 22, 26, 28, 30, 30, 32, 34, 90. The outlier is 90.
$\text{Mean} = 292 \div 8 = 36.5$; $\text{median} = (30 + 30) \div 2 = 30$; $Q_1 = 27$ and $Q_3 = 33$, so the interquartile range is 6; $\text{range} = 90 - 22 = 68$.
Without 90: $\text{mean} = 202 \div 7 \approx 28.86$, $\text{median} = 30$, $\text{range} = 12$. The mean moves by about 7.64 and the median by 0.
True: The mean is greater than the median. Removing 90 from the data set would change the mean more than it would change the median. The range is more than twice the interquartile range.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Re-check strip (3 items, C-DP.1.1-R)

1. Regular **B. the median, because an outlier pulls the mean away from the rest**

210 is far greater than every other value, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

2. Regular **D. the mean, because no value is far from the rest of the data**

The bars form one unbroken group, and no value is set apart from the rest, so the data set has no outlier.
With no outlier to pull it off center, the mean uses every value and describes a typical value well.

3. Regular **A. the interquartile range, because one long whisker stretches the range**

The box holds the middle half of the data: $Q_3 - Q_1 = 30 - 20 = 10$ minutes.
The right whisker reaches 65, which is 35 minutes from the box, more than three times the width of the box. The highest data lie far from the middle half of the data.
Those data stretch the range, $65 - 15 = 50$ minutes, but the interquartile range describes only the middle half and is not affected by them, so the interquartile range describes the spread better.

Comparison strip (2 items, C-DP.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **A. the interquartile range, because an outlier stretches the range**

153 is far from every other value, so it is an outlier.
The range, $153 - 33 = 120$ customers, is stretched by the outlier. The interquartile range uses only the middle half of the ordered data, $Q_3 - Q_1 = 59 - 43 = 16$ customers, so the outlier does not stretch it.
The interquartile range describes the spread better; it goes with the median as the measure of center.

2. Regular **B. the median, because one value is far from the rest of the data**

One value, in the 70–79 interval, is separated from all the others by three empty intervals, so it is an outlier.
An outlier pulls the mean toward it, but the median, the middle of the ordered data, stays with the other values. The median describes a typical value better.

Reteach plan — C-DP.1.1

MA.7.DP.1.1 · Choose a measure of center or variation

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.1-R

Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____ Calculator: yes

1. The yearly salaries at a small shop, in thousands of dollars, are 26, 28, 34, 46, 52, 210.

Which measure of center best describes a typical value in this data set?

A. the mean, because no value is far enough away to pull it off center

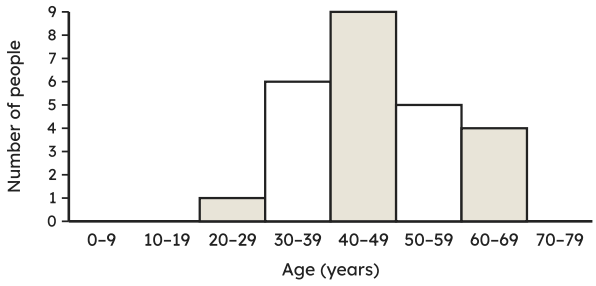
B. the median, because an outlier pulls the mean away from the rest

C. the median, because it is easier to calculate than the mean

D. the mean, because only the middle value affects it

2. The histogram shows the ages of the 25 people at a family picnic.

Which measure of center best describes a typical value in this data set, and why?



A. the mean, because one value is far from the rest of the data

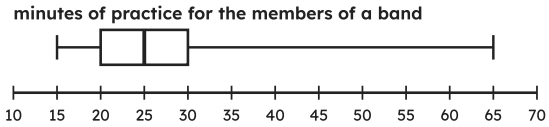
B. the range, because no value is far from the rest of the data

C. the median, because one value is far from the rest of the data

D. the mean, because no value is far from the rest of the data

3. The box plot shows the minutes of practice for the members of a band.

Which measure of variation better describes the spread of the data, and why?



A. the interquartile range, because one long whisker stretches the range

B. the median, because it is in the middle of the box

C. the range, because both whiskers are short and nothing stretches it

D. the interquartile range, because it uses every value in the data set

Score: ____ / 3 → copy to your progress strip

Benchmark: **MA.7.DP.1.1** · Choose a measure of center or variation Student code: _____ Calculator: yes

1. The numbers of customers at a food truck on six days are 33, 43, 50, 55, 59, 153.

Which measure of variation better describes the spread of the data, and why?

A. the interquartile range, because an outlier stretches the range

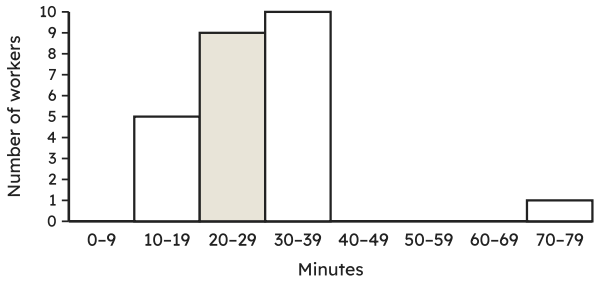
B. the range, because an outlier makes the spread easier to see

C. the range, because there is no outlier to stretch it

D. the median, because it is the middle value of the ordered data

2. The histogram shows the commute times, in minutes, of 25 workers at an office.

Which measure of center best describes a typical value in this data set, and why?



A. the range, because one value is far from the rest of the data

B. the median, because one value is far from the rest of the data

C. the mean, because no value is far from the rest of the data

D. the mean, because one value is far from the rest of the data

Score: ____ / 2 → copy to your progress strip